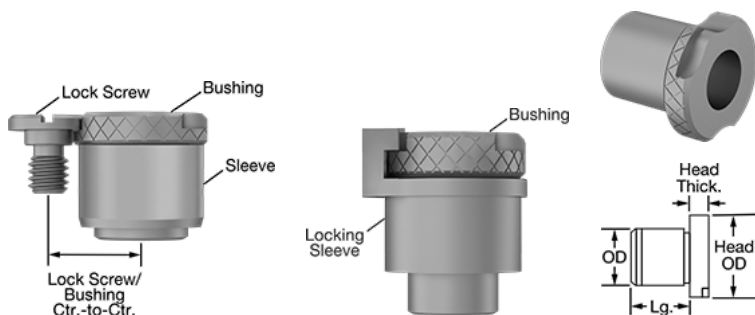


Removable Drill Bushings



Replace worn bushings and quickly change IDs without needing to repeatedly use a lev bushings in and out of your jig fixture, which can cause damage, these bushings slide i separately). By swapping in a bushing with a different ID, you can change the size of y hole in your fixture. All are made to tight dimensional tolerances for consistent, accur change them in and out. They're also known as slip-fixed and renewable bushings.

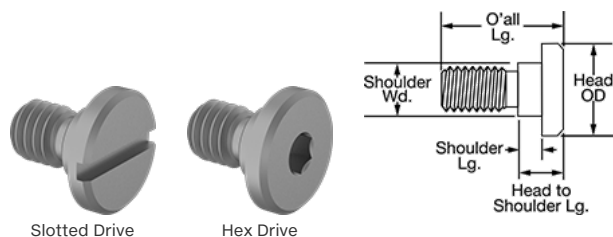
Choose a sleeve with a length that matches the length of your bushing. To prevent bust the sleeve as you drill, pair a standard sleeve with a lock screw, or use a locking sleeve removal from sleeves, use a [handle](#).

		Head			Lock Screw/Bushing			Tolerance			Drill Bushing		
OD	Lg.	For Drill Bit Size	For Drill Bit Size Decimal Equiv.	OD	Thk.	Lock Screw/Bushing Ctr.-to-Ctr.	ID	OD	Lg.	Type	Material	Hardness	
0.25" ID													
3/4"	1/2"	1/4", E	0.25"	1 3/64"	1/4"	3/4"	0.0001" to 0.0004"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A0
0.29" ID													
3/4"	1"	L	0.29"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.3125" ID													
3/4"	1/2"	5/16"	0.3125"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	3/4"	5/16"	0.3125"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	1"	5/16"	0.3125"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.315" ID													
3/4"	1"	8 mm	0.315"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.323" ID													
3/4"	1"	P	0.323"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.3281" ID													
3/4"	2 1/8"	21/64"	0.3281"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A0
0.339" ID													
3/4"	1"	R	0.339"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.3438" ID													
3/4"	1 3/8"	11/32"	0.3438"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.348" ID													
3/4"	1"	S	0.348"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.3594" ID													
3/4"	1 3/8"	23/64"	0.3594"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.368" ID													
3/4"	1"	U	0.368"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.375" ID													
3/4"	1/2"	3/8"	0.375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	3/4"	3/8"	0.375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	1"	3/8"	0.375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.376" ID													
3/4"	1"	3/8"	0.376"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.377" ID													
3/4"	1"	V	0.377"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.386" ID													
3/4"	1"	W	0.386"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.3906" ID													
3/4"	1/2"	25/64"	0.3906"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	3/4"	25/64"	0.3906"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	1"	25/64"	0.3906"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.397" ID													
3/4"	1"	X	0.397"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.404" ID													
3/4"	1"	Y	0.404"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.4063" ID													
3/4"	1/2"	13/32"	0.4063"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	3/4"	13/32"	0.4063"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.413" ID													
3/4"	1 3/8"	Z	0.413"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5
0.4219" ID													
3/4"	1/2"	27/64"	0.4219"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	3/4"	27/64"	0.4219"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	1"	27/64"	0.4219"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.4375" ID													
3/4"	3/8"	7/16"	0.4375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A0
3/4"	1/2"	7/16"	0.4375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	3/4"	7/16"	0.4375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	1"	7/16"	0.4375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
3/4"	1 3/8"	7/16"	0.4375"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2
0.4531" ID													
3/4"	1"	29/64"	0.4531"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A3

Head								Tolerance			Drill Bushing			
OD	Lg.	For Drill Bit Size	For Drill Bit Size Decimal Equiv.	OD	Thk.	Lock Screw/Bushing Ctr.-to-Ctr.	ID	OD	Lg.	Type	Material	Hardness		
0.4688" ID														
3/4"	1/2"	15/32"	0.4688"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	3/4"	15/32"	0.4688"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	1"	15/32"	0.4688"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	1 3/4"	15/32"	0.4688"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
0.4844" ID														
3/4"	1 3/8"	31/64"	0.4844"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.01" to 0.01"	SF	Steel	Rockwell C61	8493A5	
0.5" ID														
3/4"	3/8"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A3	
3/4"	1/2"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	5/8"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A3	
3/4"	3/4"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	1"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	1 3/8"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	1 1/2"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A0	
3/4"	1 3/4"	1/2"	0.5"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A3	
0.5156" ID														
3/4"	1"	33/64"	0.5156"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A0	
0.5313" ID														
3/4"	1/2"	17/32"	0.5313"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
3/4"	3/4"	17/32"	0.5313"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A2	
0.5625" ID														
3/4"	3/4"	9/16"	0.5625"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A3	
3/4"	1"	9/16"	0.5625"	1 3/64"	1/4"	3/4"	0.0001" to 0.0005"	-0.0002" to 0"	-0.015" to 0.015"	SF	Steel	Rockwell C61	8493A3	

Accessories for Bushings

Drill Bushing Lock Screws



Lock screws thread into a hole in your jig. The head of the screw fits into a notch on the bushing's head keeps the bushing from spinning.

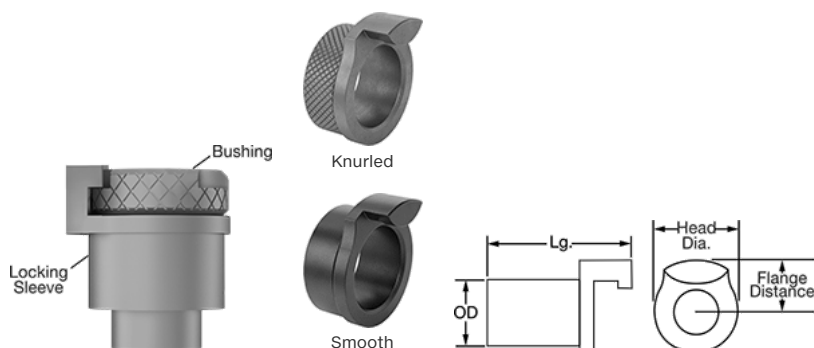
For Bushing OD	Thread Size	Head OD	Overall Lg.	Head-to-Shoulder Lg.	Shoulder		Hex Size	For Drill Bushing Type	Material
					Lg.	Wd.			
Slotted Drive									
5/16", 1/2", 3/4"	5/16"-18	5/8"	5/8"	1/4"	0.138"	3/8"	—	SF	Steel
Hex Drive									
5/16", 1/2", 3/4"	5/16"-18	5/8"	5/8"	1/4"	0.138"	3/8"	3/16"	SF	Steel

Drill Bushing Sleeves



For Bushing OD	ID	OD	Lg.	ID Tolerance	For Drill Bushing Type	Material	Hardness	Sleeve ANSI No.	Each
3/4"	3/4"	1"	3/8"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	L-64-6	3556N179 \$11.26
3/4"	3/4"	1"	1/2"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	L-64-8	3556N181 10.39
3/4"	3/4"	1"	3/4"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	L-64-12	3556N182 10.39
3/4"	3/4"	1"	1"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	L-64-16	3556N183 11.01
3/4"	3/4"	1"	1 3/8"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	L-64-22	3556N184 11.91
3/4"	3/4"	1"	1 3/4"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	L-64-28	3556N185 14.63

Drill Bushing Lock Sleeves



Lock sleeves act as a sleeve and lock screw at the same time. They have a raised flange that fits into the notch on the bushing's head, keeping the bushing in place.

Knurled—Knurled sleeves grip the sides of a hole to prevent rotation. Press or mold them into plastic tooling.

Smooth—Smooth sleeves are slightly oversized for a snug fit. They're usually installed in metal tooling—press them into place with a manual or hydraulic press.

For Bushing OD	ID	OD	Lg.	Head Dia.	Flange Distance	ID Tolerance	For Drill Bushing Type	Material	Hardness	Knurled		Smooth	
										Each		Each	
3/4"	3/4"	1"	3/4"	1 1/16"	21/32"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	31425A128	\$43.30	31425A115	\$53.14
3/4"	3/4"	1"	1"	1 1/16"	21/32"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	31425A129	49.38	31425A116	56.49
3/4"	3/4"	1"	1 1/4"	1 1/16"	21/32"	0.0003" to 0.0006"	SF	Steel	Rockwell C61	31425A131	56.05	31425A117	72.65

Removable Drill Bushings for Thin Metal